WEB-BASED THESIS ARCHIVING SYSTEM

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RATIONALE

A Thesis Archiving System is a web-based platform that allows students and faculty of St. Cecilia's College Cebu, Inc. to easily store, manage, and access academic research. Currently, theses are handled manually, making them difficult to locate and prone to loss or damage. To solve this problem, the proposed system will centralize and digitize research documents, improving accessibility, organization, and long-term preservation.

SUPPORT LITERATURE

(Shiweda, 2018) conducted a study on the usability of digital institutional repositories by postgraduate students and faculty. The study found that institutional repositories are seen to be generally useful by students and faculty, although they depend on awareness and institutional support to engage in practice.

SUPPORT LITERATURE

(Raina et al., 2023) stated that metadata standards are essential to research data repositories because they facilitate discoverability, preservation, and reuse of archived materials, and descriptive, administrative, and structural metadata facilitate key preservation and access functions.

SIGNIFICANCE OF THE STUDY

- Students
- Research Advisers
- School Administration
- Researchers
- Future Researchers

OBJECTIVES

GENERAL OBJECTIVE

To develop a web-based Thesis Archiving System for the students and faculty of St. Cecilia's College – Cebu, Inc. that will improve the storage, organization, and retrieval of academic research, and provide a more convenient and accessible platform for managing and accessing thesis documents.

SPECIFIC OBJECTIVE

- Implement Role-Based Access Control (RBAC) to ensure that students, faculty advisers, department coordinators, and administrators only have access to items and authored theses that can be relevant to their delegated roles.
- Implement a secure authentication and authorization mechanism to prevent unauthorized access, alteration or loss of thesis records.

SPECIFIC OBJECTIVE

- Offer a systematized online submission system through which the students can submit soft copies of their approved theses as official online backup of the hardbound version.
- Maintain document integrity and consistency by implementing file format validation and uploading limitations to allow only authorized manuscript file formats (like PDF to serve as the primary archival format and DOCX to serve as revision and review) to be uploaded.

SPECIFIC OBJECTIVE

- Implement a version control mechanism to record changes made to thesis submissions, ensure that historical data of updated documents is kept to ensure transparency and accountability.
- Integrate structured metadata management, including thesis title, abstract, keywords, author, department, adviser, year, and related information, to harmonize the records and enhance searchability in the repository.

SPECIFIC OBJECTIVE

- Implement an audit trail system to track user activities such as logins, uploads, downloads, and updates to ensure accountability and system security.
- Develop a monitoring dashboard to be used by administrators to access system statistics such as the total number of theses uploaded, documents pending submission, approved documents, user activity, and system performance.

SPECIFIC OBJECTIVE

- Create a monitoring dashboard to be used by administrators to access system statistics such as the total number of theses uploaded, documents pending submission, approved documents, user activity, and system performance.
- Establish backup and recovery facilities at the system level to safeguard the uploaded thesis files and other data against accidental or corruption or system failure.

SPECIFIC OBJECTIVE

- Implement status tracking to monitor thesis progress tracking in real time.
- Establish a searchable and categorized digital library that has sophisticated search filters to make sure that archived theses are easily accessed.

SCOPE AND LIMITATION

SCOPE

- **User Role and Access:** The intended users are the students, faculty advisers, research coordinators, department coordinators, and administrators. The roles have limitations, which prevent access to appropriate functions and documents.
- **Thesis Submission:** Copies of approved theses can be digitally submitted by students, and the system would monitor their status and progress on a case-by-case basis.

SCOPE

- **Thesis Management:** Research advisers and coordinators can review and provide feedback as well as monitor compliance with the institutional guidelines
- **Digital Archiving:** The system ensures that the theses are safely stored, it has versioning control of the theses, manages its metadata, and offers advanced searching and categorizing of the theses stored.

SCOPE

- **Monitoring and Reporting:** Administrators have access to dashboards whereby they can monitor the rates of submissions, pending documents, approved theses and the activities of the users and the general functioning of the system.
- **Security Measures:** The backup/recovery, audit trail, and authentication are implemented to guarantee the integrity of the data and to avoid unauthorized access to it.

SCOPE

- **System Availability:** The system is accessible via web browser.

LIMITATION

- **Physical Copies:** This system does not eliminate the necessity of having a physical copy of the thesis; it is merely a digital backup.
- **Restricted Access:** Only internal users are allowed to view or manipulate theses; external institutions are not allowed to view or modify the thesis.

LIMITATION

- **Internet Dependence:** The system needs a stable internet connection, it does not support offline submission or access.
- **File Format Constraints:** Supported file format types include PDF for archiving, DOCX for review/revision purposes, whereas all other file formats are not supported.

LIMITATION

- **No Plagiarism Detectors:** The system does not check the quality and originality of the content.
- **Scope of Feedback:** The feedback with the research advisor is obtained in the form of online comments, and it does not offer auto- evaluation and grading options.

SOFTWARE APPROACH

STRCUTURED ANALYSIS & DESIGN

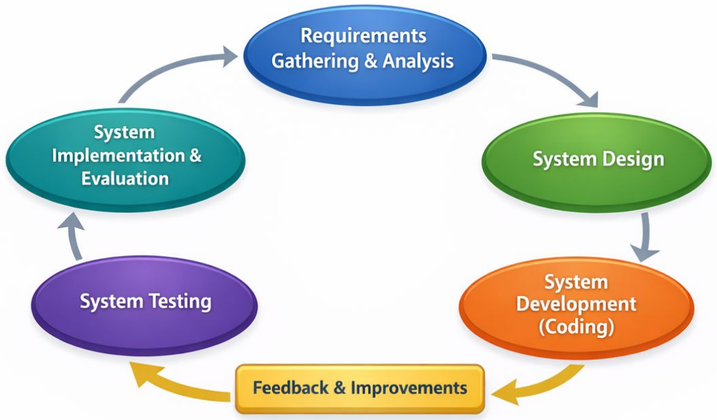


Figure 3.1 *Agile Software Development Methodology*

TOOLS & TECHNOLOGIES

- Frameworks
- Programming Languages
- API’s
- Support Technologies

SYSTEM USERS / ACTORS

Student

Research Adviser

Research Coordinator

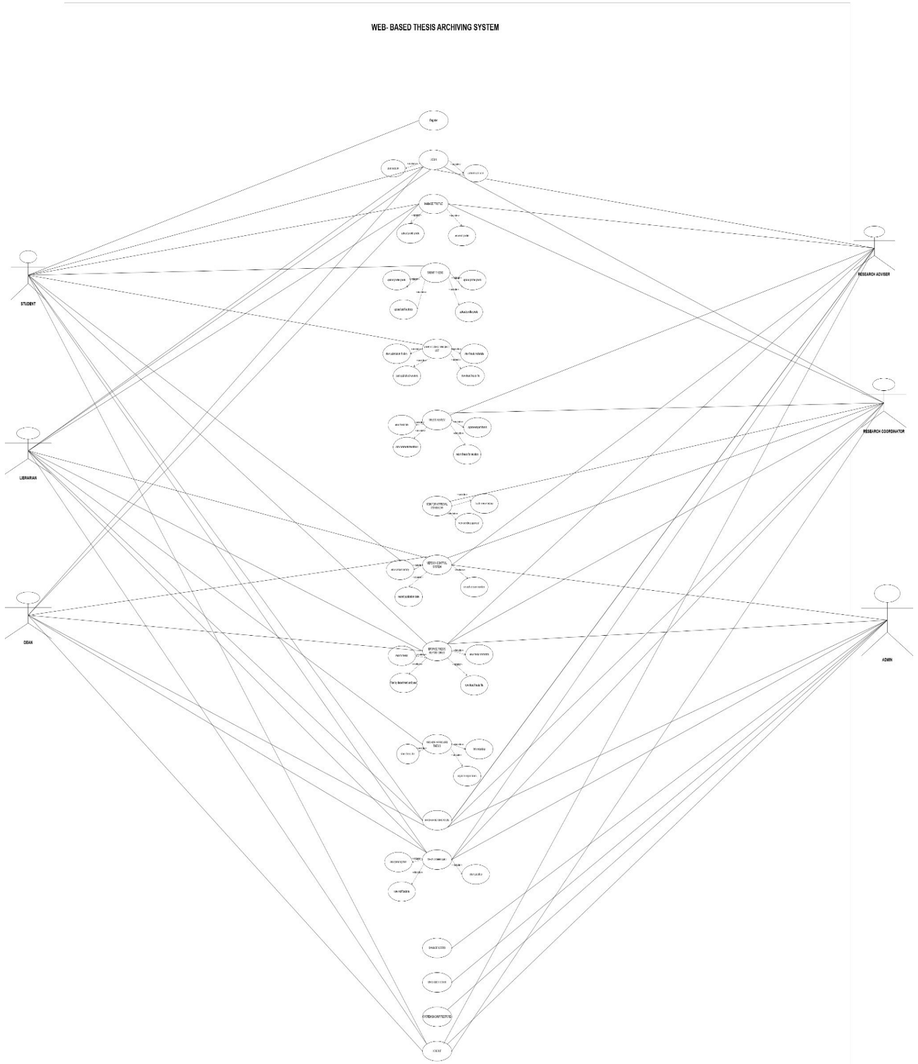
SYSTEM USERS / ACTORS

Dean

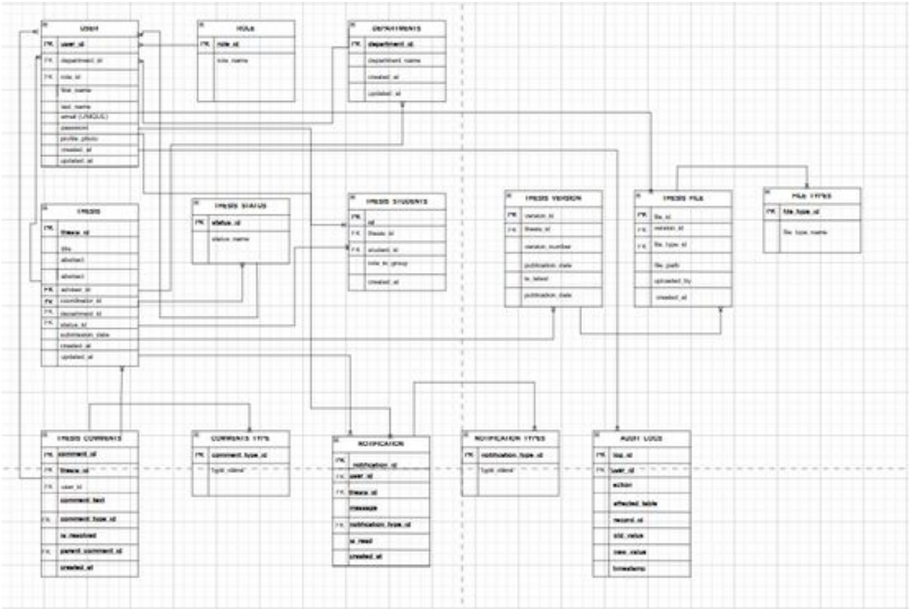
Librarian

Developer

USE-CASE DIAGRAM



ENTITY-RELATIONSHIP DIAGRAM-(ERD)



THANK YOU